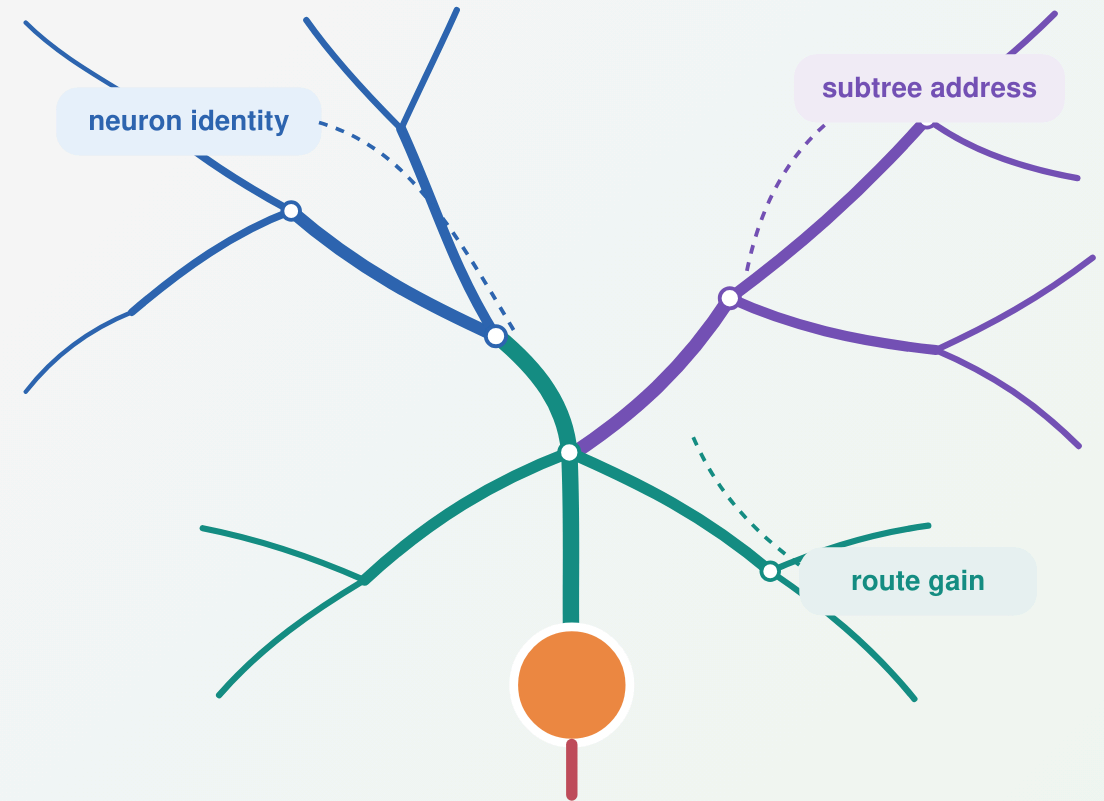


When can dendritic structure help local credit assignment?

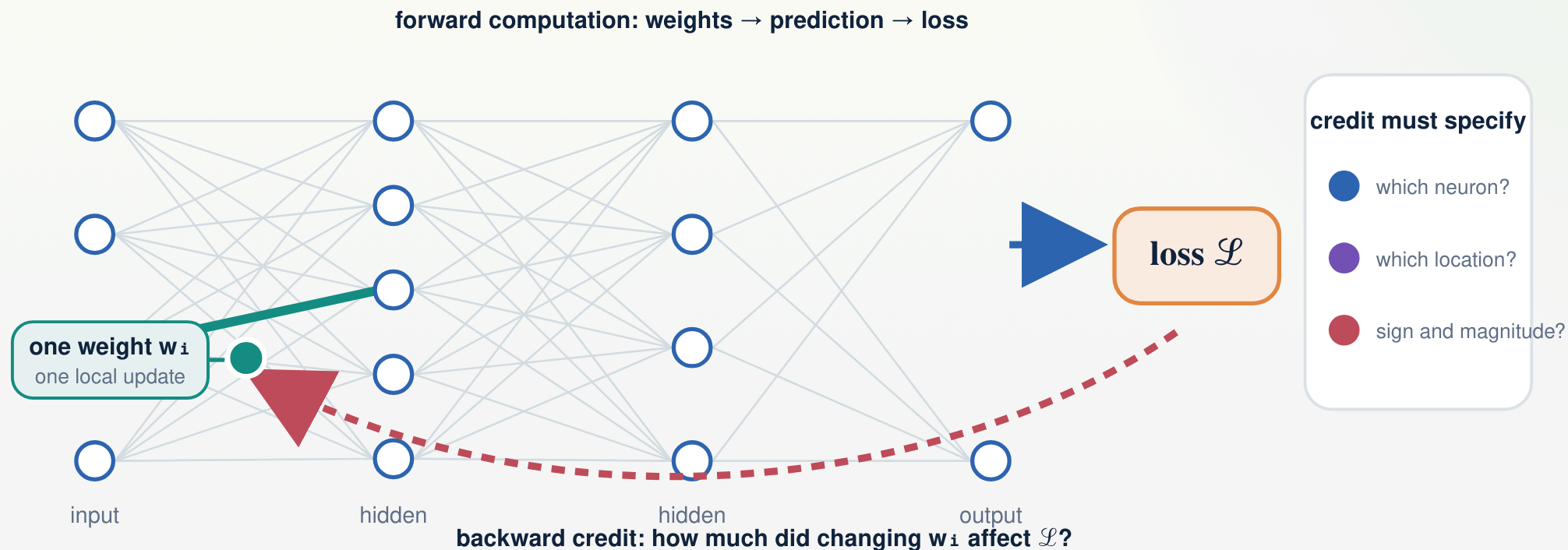
Feedback bandwidth, within-neuron address, conductance-dependent route gain, and an alignment boundary

Houman Safaai · Maceo Richards · Bernardo L. Sabatini

Kempner Institute, Harvard University · Harvard Medical School



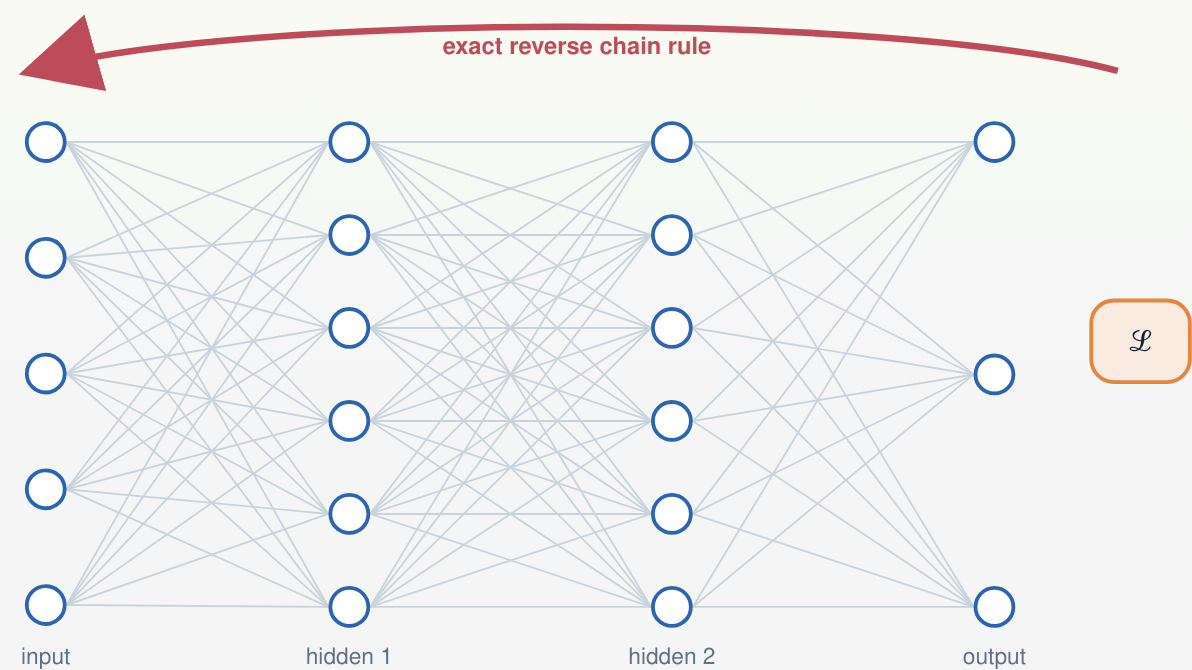
Learning changes network weights; credit assigns each change



$$\mathcal{L} = \mathcal{L}(f(x; w), y) \quad \cdot \quad \Delta w_i = -\eta \partial \mathcal{L} / \partial w_i$$

→ A global objective must be converted into a destination, sign, and magnitude for every trainable weight.

Backpropagation defines the exact neuron-specific learning signal



$$z_u = \sum_{i \in \text{in}(u)} w_{i,u} x_i + b_u, \quad y_u = f_u(z_u)$$

$$\delta_u \equiv \partial \mathcal{L} / \partial y_u = \sum_{v: u \rightarrow v} w_{u,v} f'_v(z_v) \delta_v$$

δ_u is exact task information assigned to neuron u —not yet to one weight.

fixed/random feedback	feedback alignment; direct feedback
inferred targets or states	equilibrium propagation; predictive coding
eligibility + learning signal	three-factor rules; e-prop
dendritic teaching signals	somato-dendritic and apical-error models

➔ Backpropagation is the information reference; biological theories differ in how the returned signal is produced.

Local learning preserves eligibility and approximates the returned signal

$$\partial \mathcal{L} / \partial w_i = \underbrace{x_i f'_u(z_u)}_{\text{local eligibility } e_i} \times \underbrace{\delta_u}_{\text{exact neuron signal}}$$

$$\Delta w_i = -\eta e_i \delta_u^{\text{avail}}$$

one layer-wide scalar m

Every neuron receives the same task-dependent coordinate.

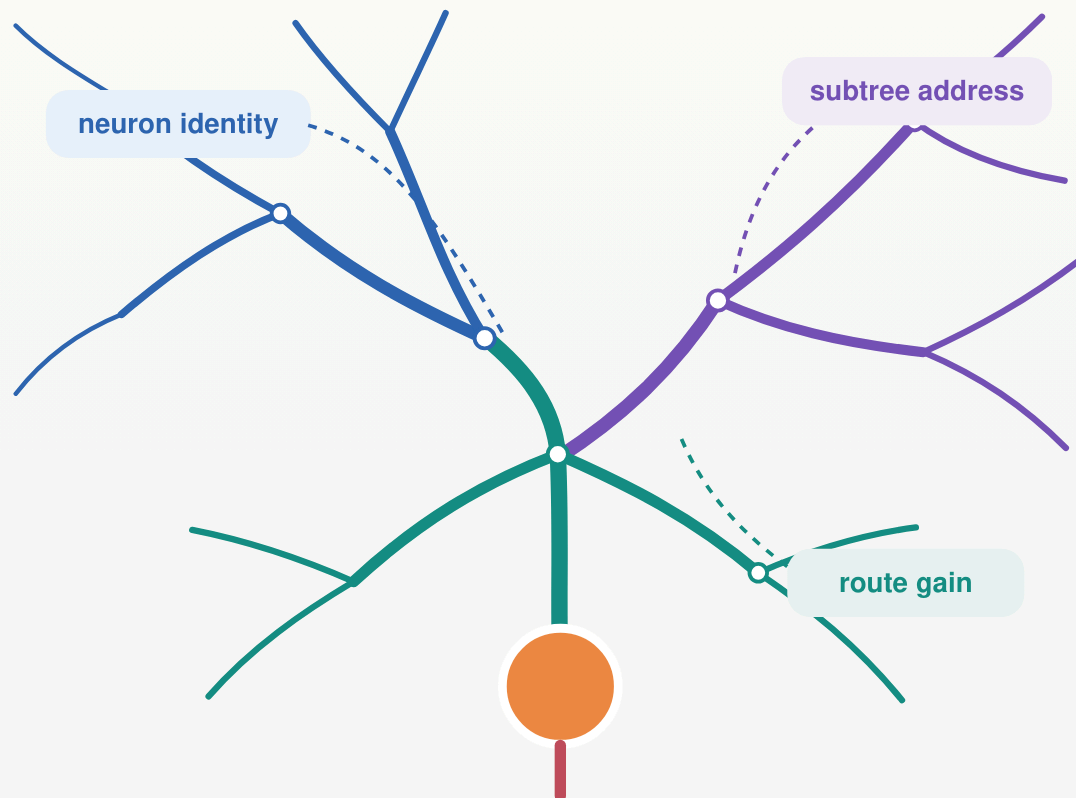
one signal δ_u^{avail} per neuron

Feedback identifies the postsynaptic neuron while the eligibility remains weight-specific.

A shared scalar does not equalize the weights: each update still contains a different eligibility e_i .

→ **Locality determines where an update is computed; feedback bandwidth determines what task information it can use.**

Dendrites add within-neuron state, address, and route gain



network weight $w_1 \rightarrow$ synaptic conductance g_1 on compartment n

$$\delta V, \text{avail}_u = A_u \beta_u, \quad A_u \in \mathbb{R}^{N_u \times K}$$

N_u : compartments · K : independent within-neuron signals · column k of A_u : route- k support and gain

local state

Voltage and conductance determine the eligibility at each synapse.

subtree address

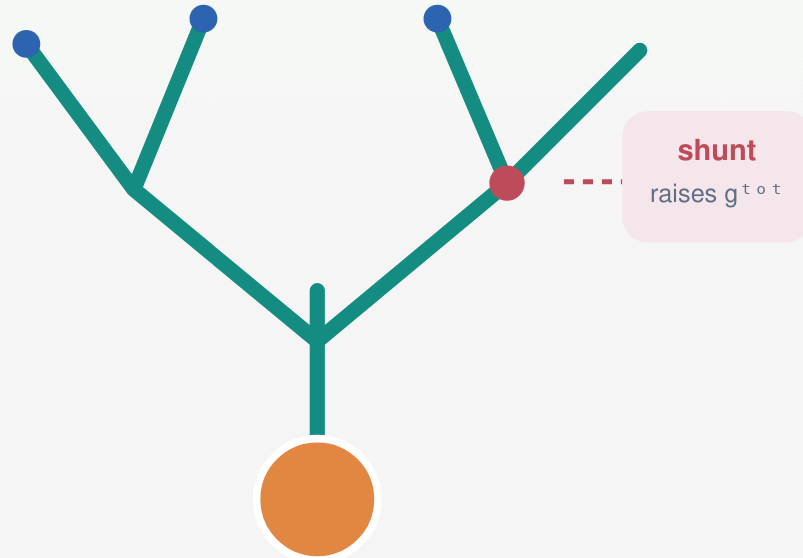
A few coefficients β_u can target nested groups of synapses.

route gain

Conductance can scale how strongly a returned signal reaches one path.

→ Neuron identity and within-neuron address are distinct credit-assignment problems.

Conductance makes dendritic voltage a normalized quotient



$$g_n^{\text{tot}} = g_n^L + \sum_{i \in \text{syn}(n)} g_i x_i + \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}}, \quad R_n^{\text{tot}} = 1/g_n^{\text{tot}}$$

$$V_n = R_n^{\text{tot}} [g_n^L E^L + \sum_{i \in \text{syn}(n)} g_i x_i E_i^{\text{rev}} + \sum_{c \in \text{child}(n)} g_{c \rightarrow n}^{\text{den}} a_c], \quad a_c = f_c(V_c)$$

additive current

Changes the numerator without changing total conductance.

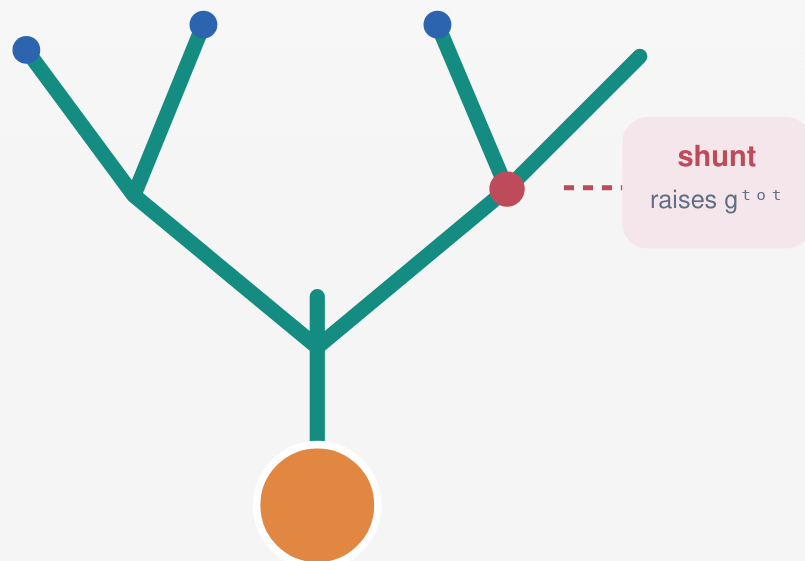
shunting conductance

Raises g_n^{tot} and lowers R_n^{tot} , even when net shunt current is small.

$V_n \approx E_I \Rightarrow g_I x_I (E_I - V_n) \approx 0$, but $g_I x_I$ still changes the denominator.

→ Shunting changes local sensitivity because conductance appears in the voltage denominator.

Dendritic credit still factorizes into eligibility $\times$ learning signal



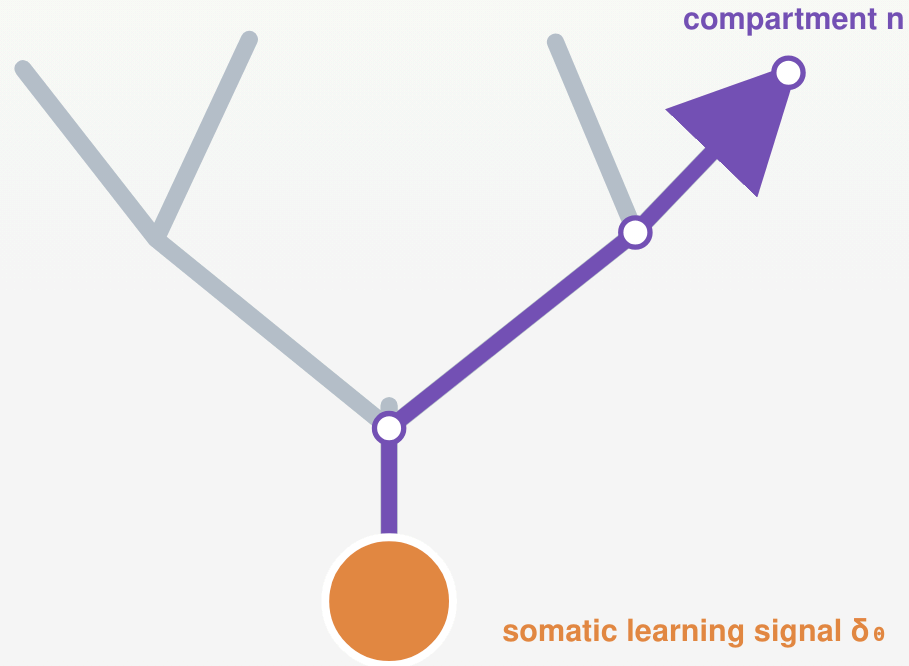
$\delta^V_n \equiv \partial \mathcal{L} / \partial V_n$ is the compartment learning signal

$$\partial V_n / \partial g_i = x_i R_n^{tot} (E_i^{rev} - V_n)$$

$$\partial \mathcal{L} / \partial g_i = \underbrace{x_i R_n^{tot} (E_i^{rev} - V_n)}_{\text{local dendritic eligibility } e_i^{den}} \times \underbrace{\delta^V_n}_{\text{returned compartment signal}}$$

→ The local factor changes with dendritic state; the circuit-level learning signal still has to reach the correct compartment.

Tree transport turns one somatic signal into a compartment field



$$\delta V_0 = f'_0(V_0)\delta u, \quad \delta V_n = \delta V_0 \gamma_n$$

$$\gamma_n = \prod_{(j \rightarrow k) \in \text{path}(n \rightarrow 0)} f'_j(V_j) R_k^{\text{tot}} g_{j \rightarrow k}^{\text{den}}$$

directed tree

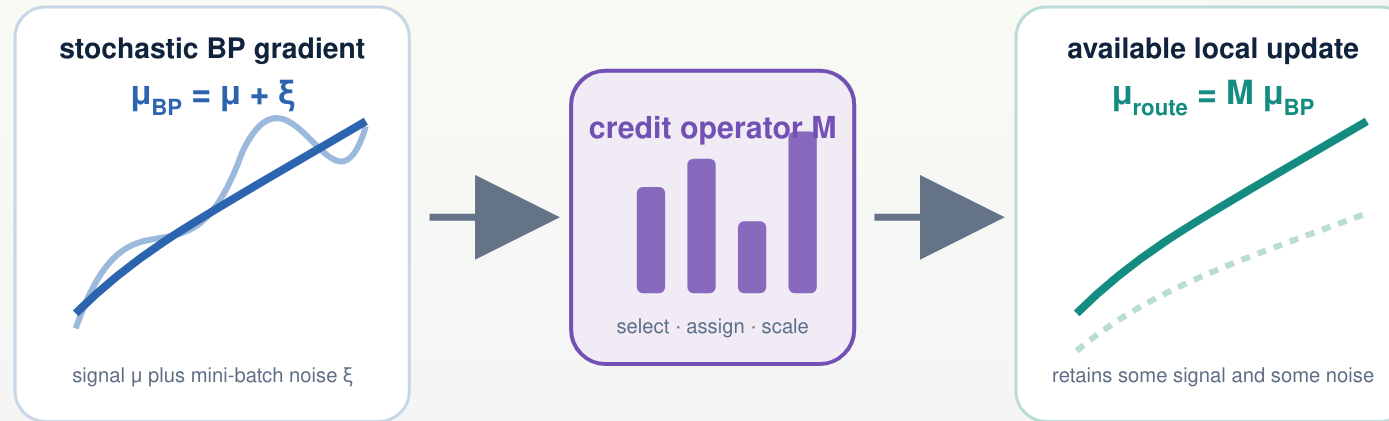
The product is indexed child $\rightarrow$ parent ($n \rightarrow 0$); returned credit propagates over the same path in reverse ($0 \rightarrow n$).

reconstructed reciprocal cable

The corresponding field is obtained from the steady-state adjoint $J_V^T q = \nabla_V \mathcal{L}$.

➔ Dendrites do not generate the task error; they transform a neuron-level signal into spatially structured compartment credit.

A feedback pathway selects, assigns, and scales stochastic credit



$$\mu_{BP} = \mu + \xi, \quad E[\xi] = 0, \quad \text{Cov}(\xi) = \Sigma$$

$$\mu_{route} = M \mu_{BP}, \quad w += w - \eta \mu_{route}$$

$M = I$: unrestricted backpropagation of the stochastic gradient. Restricted routes change the span, assignment, or gain of the available update.

→ The operator M lets one theory describe scalar, neuron-specific, subtree-targeted, and exact compartment feedback.

Operator utility balances retained signal, update cost, and noise

OPERATOR UTILITY · POSITIVE TASK ALIGNMENT REQUIRED

$$U(M) = \frac{[\mu^T M \mu]^2}{2L_{sm}[\|M\mu\|^2 + \text{tr}(M\Sigma M^T)]}$$

task-aligned signal

finite-step update cost

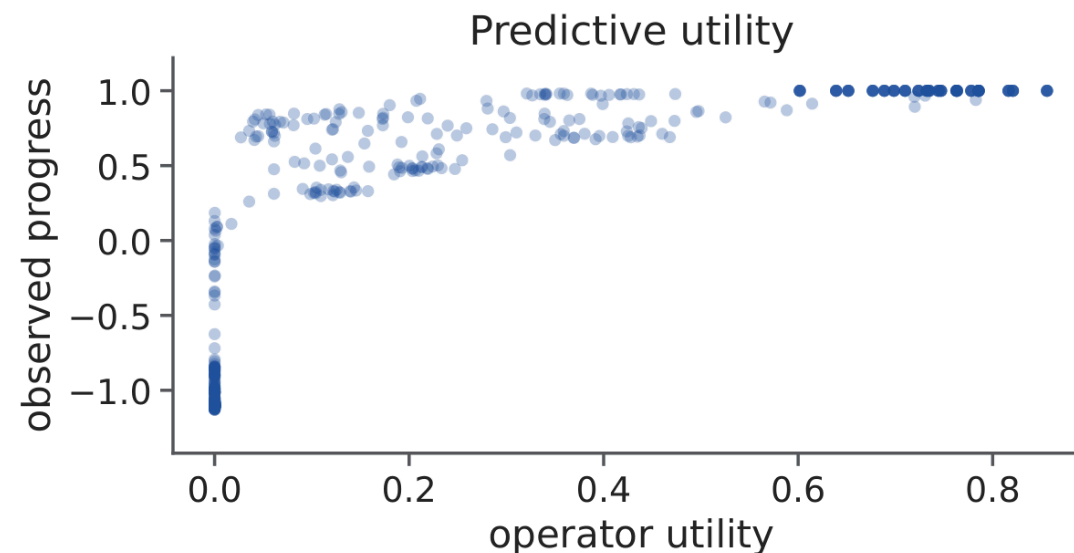
admitted stochastic noise

$\rho_s = 0.937$

utility versus observed norm-matched one-step progress across 540 trained conditions

$\rho_s = 0.916$

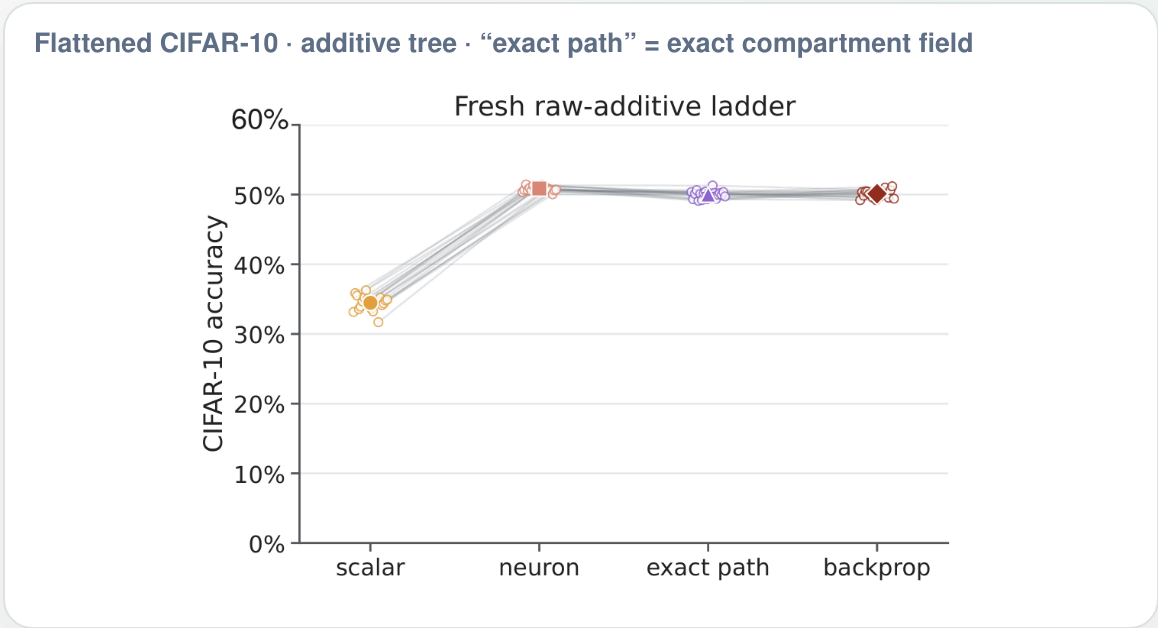
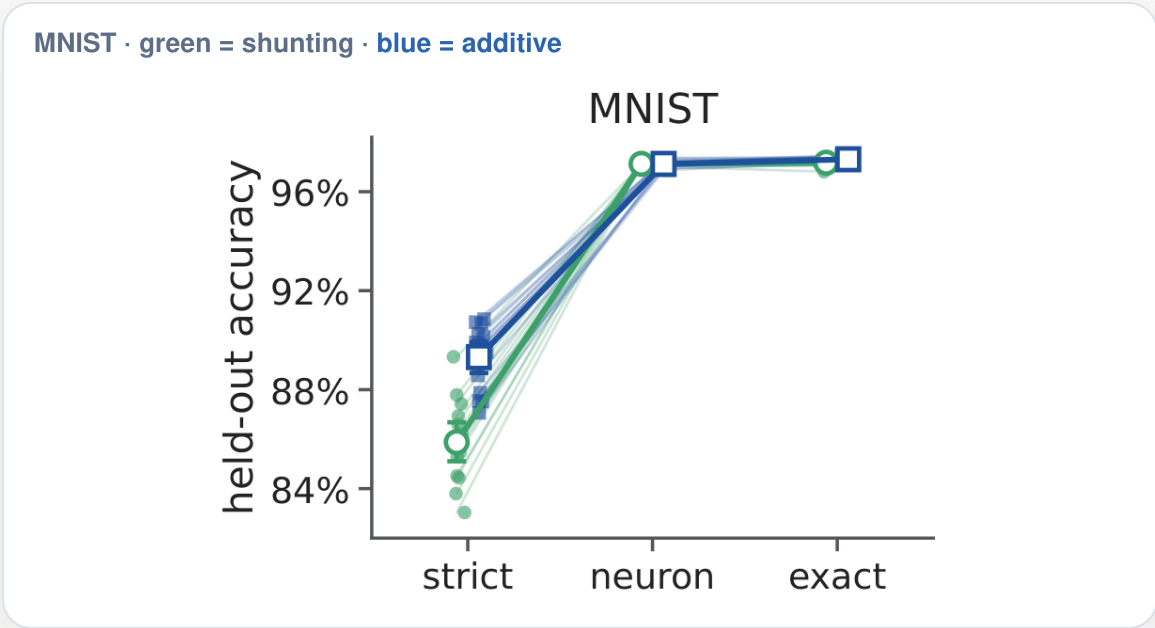
utility versus final trained accuracy



Exact for an isotropic quadratic with Hessian $L_{sm}I$; otherwise a curvature bound and one-step smoothness guarantee. It predicts large contrasts and within-family progress—not every trajectory-accrued difference.

➔ Restricted routing helps only when its retained, aligned signal compensates for the signal it discards and the noise it admits.

Neuron identity—not exact path resolution—dominates both standard tasks

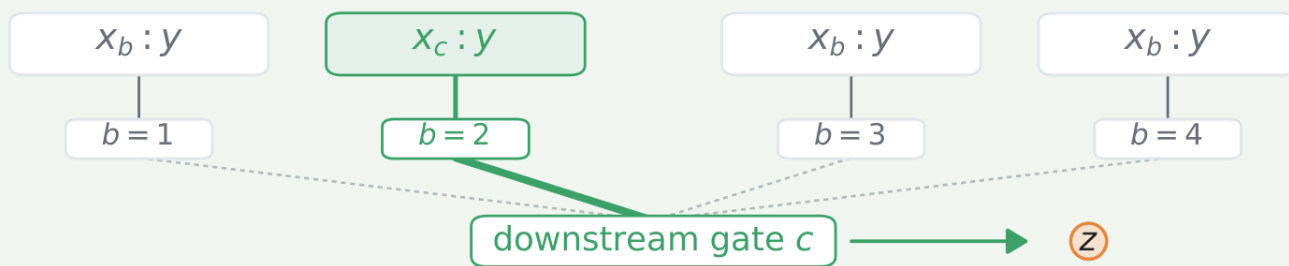


model/task	scalar → neuron	neuron → exact field
MNIST · shunting	+11.25 pp	+0.045 pp
MNIST · additive	+7.81 pp	+0.186 pp
CIFAR-10 · additive	+16.39 pp	-0.86 pp

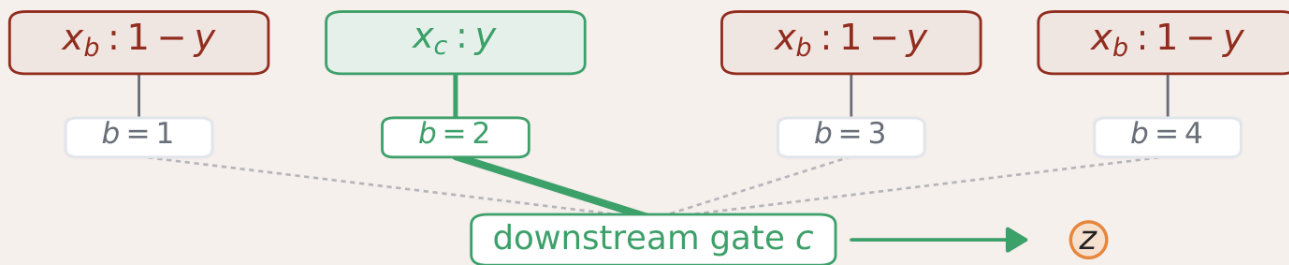
➔ On both tasks, most of the gain comes from selecting the correct neuron; exact within-tree resolution adds little.

Credit conflict creates a demand for branch-specific signals

compatible, $\chi = 0$



conflicting, $\chi = 1$



input

All B branches receive a Fashion-MNIST image and form nonzero eligibility.

forward selector

Context c selects the branch whose image determines the target.

conflict dose χ

Nonselected images vary from class-compatible ($\chi=0$) to opposite-class ($\chi=1$).

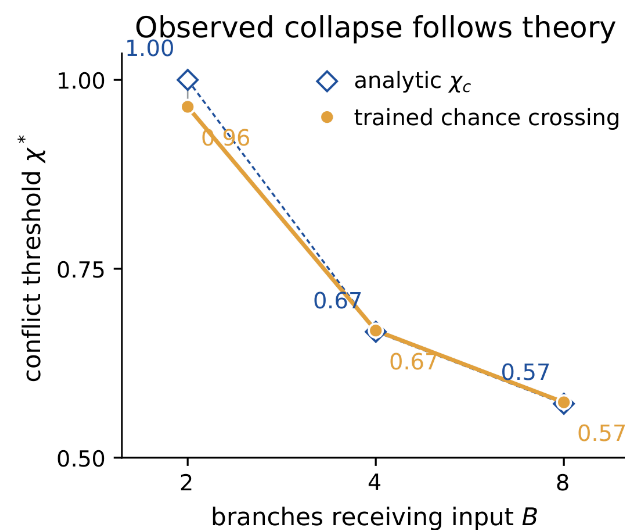
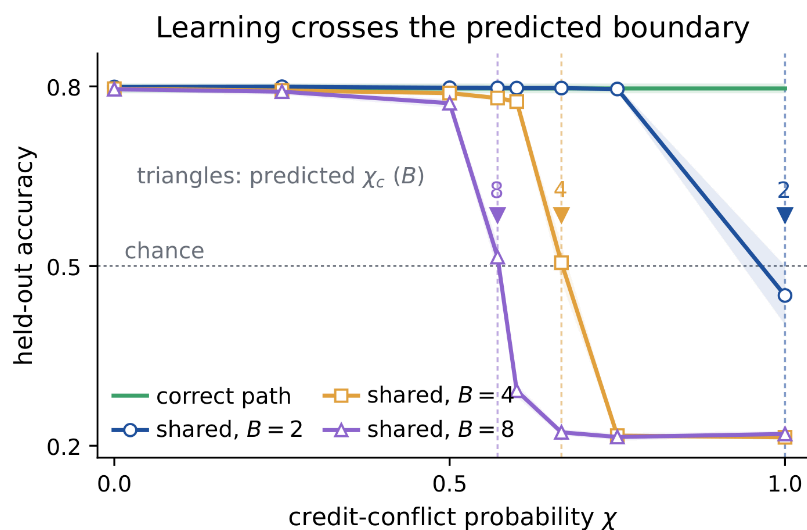
$$d^{\text{branch}}_b = N^{-1} \sum_t \delta_t \mathbf{1}[c_t=b] x_{t,b}$$

$$d^{\text{shared}}_b = N^{-1} \sum_t \delta_t B^{-1} x_{t,b}$$

N : trials · δ_t : downstream logit gradient · $\mathbf{1}[\cdot]$: indicator
 d_b : mean update direction for branch b

→ At zero conflict, one shared signal is sufficient; at high conflict, different branches require opposite updates.

Shared credit fails at the predicted conflict boundary



$$\lambda_{\text{shared}} = B - 2\chi(B-1) \rightarrow \chi_c = B/[2(B-1)]$$

35–58 pp

branch-specific gain over shared feedback at full conflict

causal control

Cyclically deranged routes fail despite identical rank and sparsity.

implementation control

Analytic BP, the correct route, and a gated-point calculation are equivalent forms of the same routing matrix.

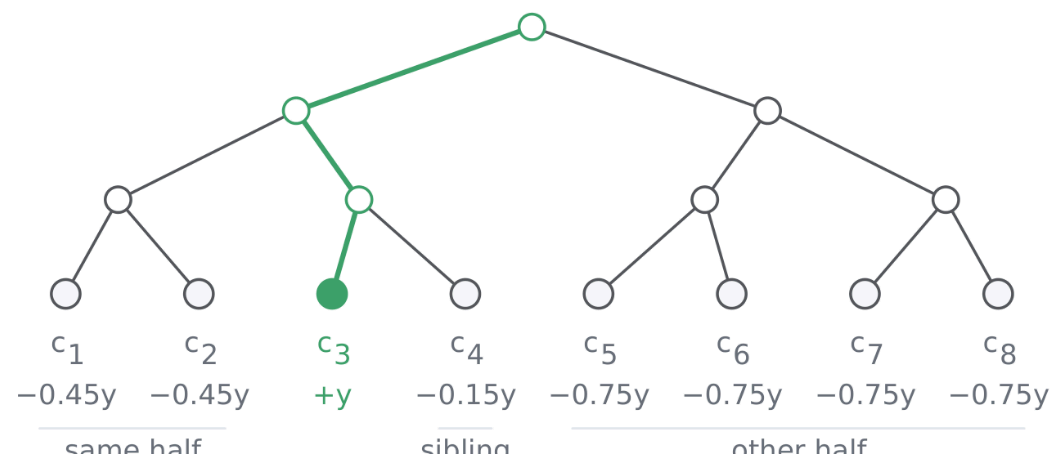
→ The task establishes a need for an address—not a unique need for dendritic material.

Nested tasks ask whether a few subtree addresses are efficient

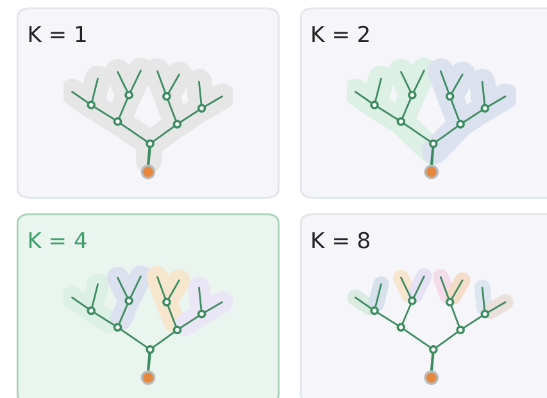
A

Eight-context hierarchical task

context $c = 3$ selects stream 3; selected stream carries $+y$; all streams active



$$\delta V, \text{avail} = A_K \beta, \quad \text{rank}(A_K) = K$$

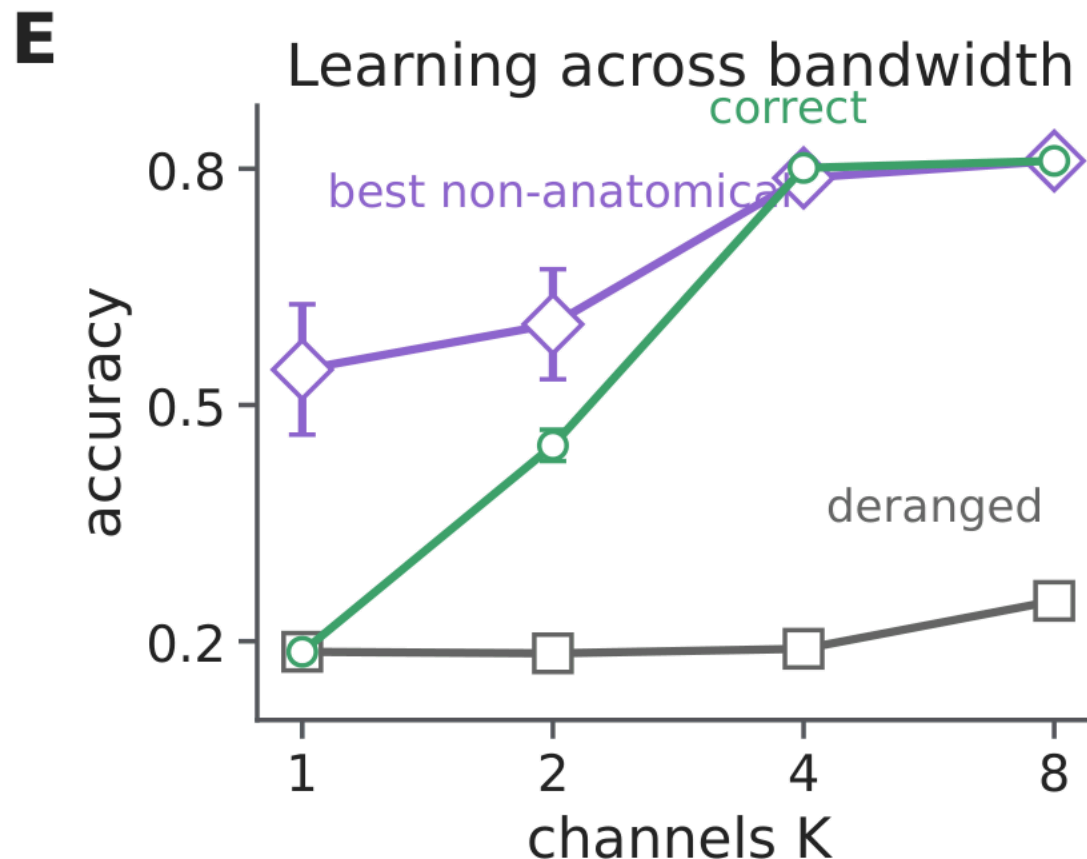
B Feedback bandwidth within one neuron


matched controls

Rank, sparsity, parameter count, and forward resources are held fixed; only route assignment, basis, or topology changes.

→ Intermediate K tests whether tree-structured feedback is a useful low-dimensional basis for task credit.

Subtree addresses help—but fine topology adds only a narrow gain



+61.1 pp

correct ancestry assignment over cyclic derangement at the same K=4 bandwidth

+1.27 pp

correct ancestry over the strongest matched non-anatomical low-rank control at K=4

against the matched low-rank basis

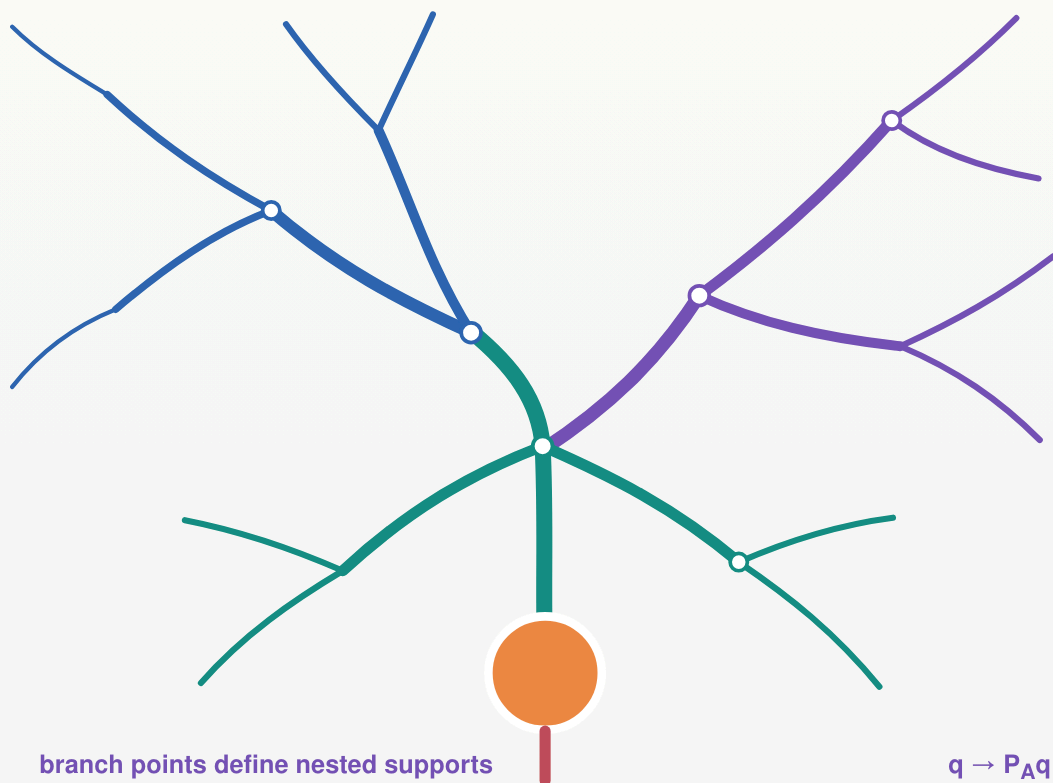
Ancestry loses at K=1,2; wins only at K=4; and ties at full rank K=8. Rewiring removes the intermediate advantage.

equivalence

Dendritic, grouped-point, and gated-point implementations coincide for the same routed field.

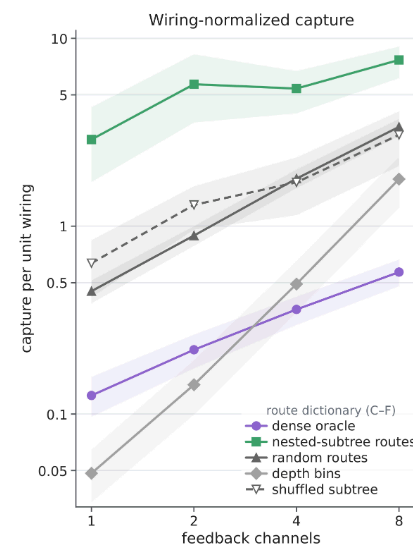
→ The large effect is the value of within-neuron addresses; the anatomy-specific effect is conditional and modest.

Real arbors provide sparse candidate routes, mostly through coarse geometry



$$C_A(q) = \|P_A q\|^2 / \|q\|^2$$

fraction of field energy in the route span



85%

of dense field capture

≈ 7%

of dense route-matrix connections

14.2×

dense capture per connection

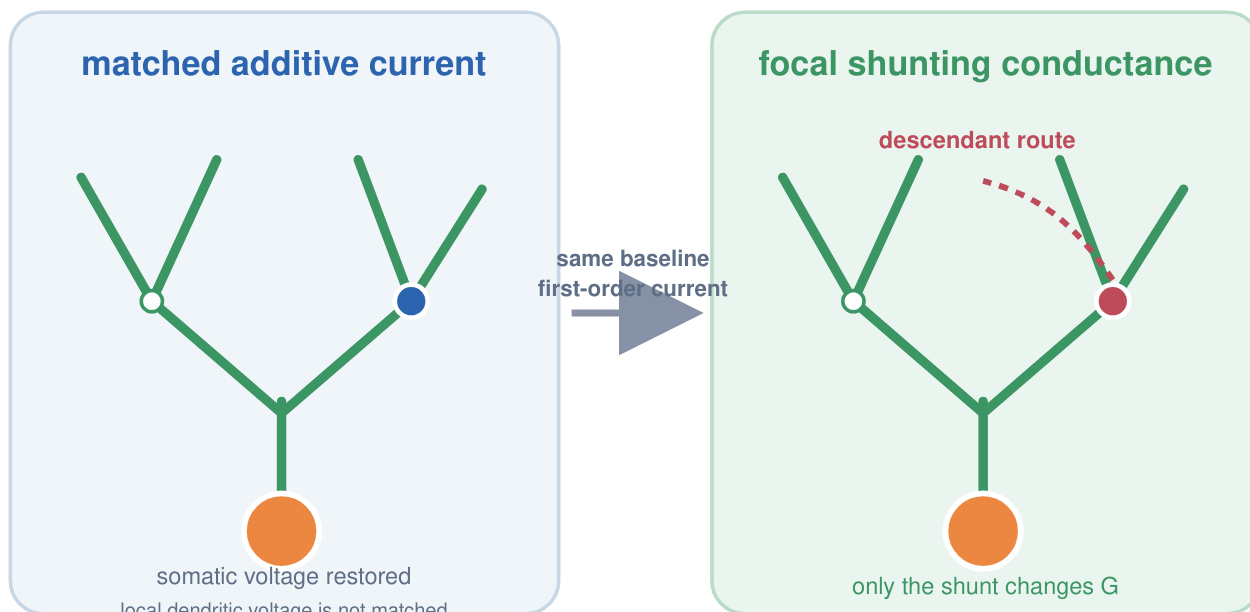
≈ 2.7×

over a density-matched shuffled dictionary

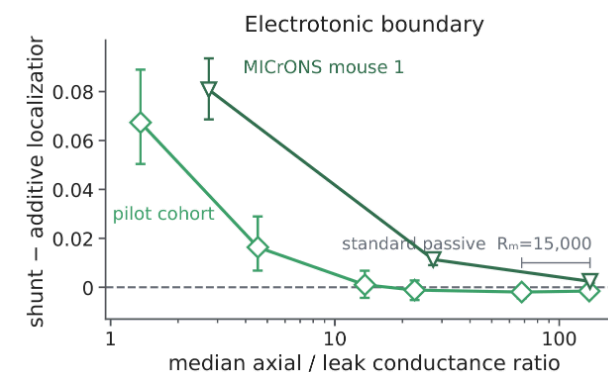
Connections are nonzero route-matrix entries—not cable length, energy, or reliability. The ordering replicated in a disjoint 47-cell cohort. A second MICrONS mouse reproduced the model-matched subtree advantage in 10/10 QC-passing cells; animal-level inference remains descriptive. Fields are modeled, not observed task gradients.

➔ Morphology supplies a sparse route dictionary; most capacity comes from branch depth and coarse topology.

Focal shunting changes descendant credit only in permissive regimes



$$q' = q - [\kappa_k q_k / (1 + \kappa_k (G^{-1})_{kk})] G^{-1} e_k$$



standard passive calibration

At $R_m = 15,000 \, \Omega \, \text{cm}^2$, the shunt-minus-current localization contrast is effectively zero.

high-conductance regime

Descendant-localized changes emerge and survive active-channel extensions.

➔ Shunting is a state-dependent route-gain mechanism, not a generic learning advantage.

Measured visual responses show no morphology-specific alignment

measured presynaptic responses



mapped conductances on each reconstructed target arbor



held-out postsynaptic-response prediction

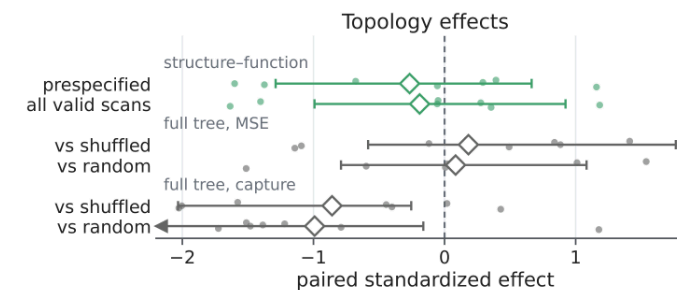
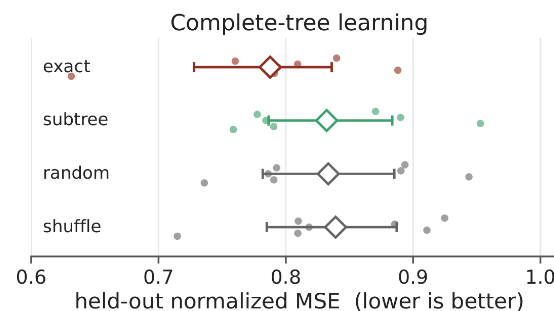
question

Does the measured input–output relation align more strongly with nested subtrees than with matched alternative route dictionaries?

inferential scope

Seven target cells from one MICrONS mouse; exact compartment errors are the reference.

Held-out learning and topology-minus-control effects



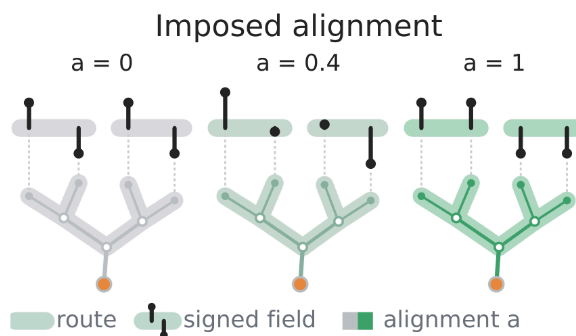
Nested subtrees do not outperform random or site-shuffled routes for held-out prediction, field capture, or within-arbor structure–function similarity.

→ Measured anatomy supplies candidate routes, but these visual responses provide no evidence that the task uses them preferentially.

The same anatomical routes capture task credit aligned to their span

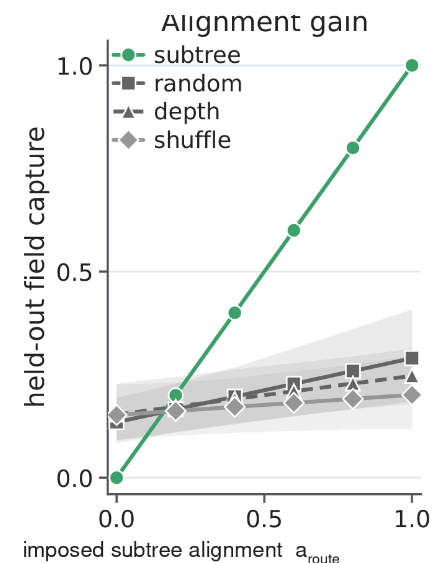
$$\varphi(a_{\text{route}}) = \sqrt{a_{\text{route}}} u_{\parallel} + \sqrt{1-a_{\text{route}}} u_{\perp}$$

$$u_{\parallel} \in \text{col}(A), \quad u_{\perp} \perp \text{col}(A), \quad \|u_{\parallel}\| = \|u_{\perp}\| = 1$$



controlled quantity

Anatomy, field energy, curvature, and route count are fixed; only alignment with the subtree span changes.



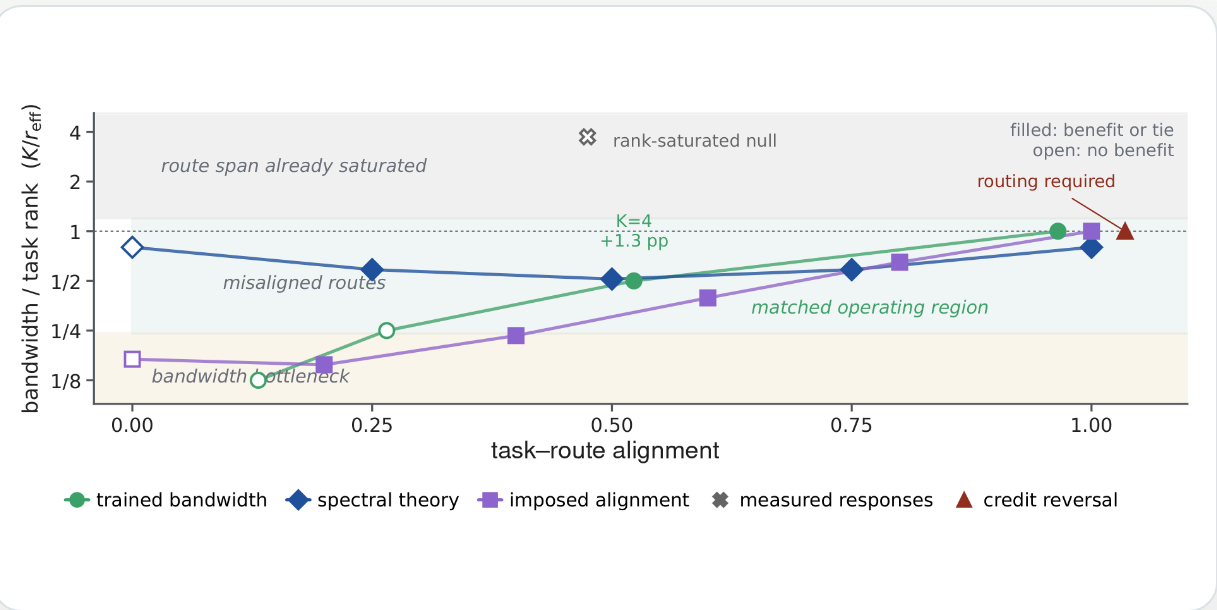
$$C_A[\varphi(a_{\text{route}})] = a_{\text{route}} \quad \text{by construction}$$

$n = 8$ reconstructed cells

Controls test whether the gain is specific to the true subtree span. The manipulation proves conditional representational sufficiency—not trained learning or endogenous biological use.

→ Alignment is sufficient for representational capture in the model; endogenous morphology-specific alignment remains unestablished.

Alignment and bandwidth determine which dendritic resources help



relative bandwidth = K/r_{eff} , $r_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2 \cdot \lambda_i$: task-credit-spectrum eigenvalues

Coordinates are estimated separately within each experiment. Regime tint and the $K/r_{\text{eff}}=1$ boundary are theoretical, not fitted.

neuron identity dominant in the standard-task model benchmarks	SUPPORTED IN MODELS
subtree addresses large when local updates conflict; topology adds a narrow gain	CONDITIONAL
route gain by shunting emerges only in permissive high-conductance regimes	CONDITIONAL
anatomical route capacity sparse candidate routes on reconstructed arbors	STRUCTURALLY SUPPORTED
endogenous morphology alignment measured visual-response analyses are null	NOT ESTABLISHED

Neuron identity usually matters first; **subtree address** helps under task-aligned conflict; **shunting** changes route gain only in permissive electrotonic states.

➔ Dendritic structure is a conditional substrate for routing local credit—not a general replacement for backpropagation.